Divide Multi-Digit Numbers Using an Algorithm

Use long division more than once in a real problem, then compare the answers.

Name _____ **Date** _____

Tonight's spark

A big school district has 621,000 students. Right now there is 1 teacher for every 20 students. The goal is 1 teacher for every 15. How many more teachers must the district hire?

Make a guess first: more than 5,000 new teachers, or fewer?

Students per teacher	Teachers
20	31,050
15	41,400
More needed	?

Our class model: two divisions, then subtract

Try it

Work it out on paper. Write one answer on each line.

1 **What is 400 ÷ 20?**

A. 2 **B.** 20 **C.** 200 **D.** 80

My answer _____

2 **A district has 8,400 students and 1 teacher for every 20 students. The division 8,400 ÷ 20 = 420 shows what?**

A. The number of teachers **B.** The number of students per teacher

C. The number of students **D.** The number of teachers to hire

My answer _____

3 **What is 1,800 ÷ 15?**

A. 12 **B.** 1,200 **C.** 120 **D.** 125

My answer _____

4 **A school has 600 students. The goal is 1 adult for every 20 students. How many adults does the school need?**

A. 20 **B.** 3 **C.** 12,000 **D.** 30

My answer _____

5 **A school has 600 students. Now there is 1 adult for every 30 students. The goal is 1 adult for every 20 students. How many MORE adults are needed?**

A. 10 **B.** 30 **C.** 20 **D.** 50

My answer _____

Talk about it

What does each quotient count, and why do you subtract at the end?

Home mission

Say the plan out loud: one teacher for every 20 students, then one for every 15. Which plan needs more teachers? How do you know before you divide?

If we get stuck, ask

- What does this quotient count — students, or adults?
- How many groups of that size fit into the total?
- Does the question ask for the new total, or for how many MORE?

Do it on a screen

Scan or type the link. Same lesson, with hints.

eduwonderlab.com/lessons/2-6-part2/homework.html

How did it feel? (optional)

Tricky Getting there Got it